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COMMERCIAL ITEM DESCRIPTION

STEREOSCOPE, LENS-PRISM-MIRROR, AERIAL PHOTOGRAPH INTERPRETATION

(WITH BINOCULARS AND CASE)

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-S-20660 for all federal agencies.

1. SCOPE. This specification covers a mirror stereoscope with binoculars and a carrying case.

2. SALIENT CHARACTERISTICS.

2.1 Description. The stereoscope is used for the examination of stereoscopic pairs of aerial photographs. The stereoscope shall consist of lenses, prisms, mirrors, binoculars, and folding legs. The binoculars are used with the stereoscope to magnify portions of the aerial photograph for closer examination.

2.2 Materials. The contractor shall select the materials, but the materials shall be capable of meeting all of the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent possible.

2.2.1 Deterioration prevention and control. The stereoscope shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the material is susceptible.

2.2.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion.

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AMSC N/A FSC 6675

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2.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

2.3 Performance characteristics.

2.3.1 Stereoscope. The stereoscope shall be of the mirrored type and shall be large enough to accommodate photographs up to 9" X 9" (22.9 X 22.9 cm) It shall be mounted on folding legs with at least one leg being adjustable to accommodate irregular surfaces. All glass surfaces shall be coated with a reflection reducing film.

2.3.1.1 Lenses. The resolving power of the stereoscope lenses, without the binoculars, shall be at least 153.85 lines per inch (6 lines per mm) at the center of the field of view. The lenses, exclusive of binoculars, shall have focal lengths of 11.25 inches (28.6 cm) ± 1 percent. They shall be free of any defect that adversely affects optical quality and when mounted shall be of sufficient size to permit easy viewing by operators having inter-pupilary distances from 2.05 to 2.95 inches (52 to 75 mm). The assembly shall be sealed to prevent entry of dirt and dust between the lenses and the prisms. The optical distance from the lenses to the plane of the photographs shall be such that the eyes of an observer are accommodated to a distance between eight feet (2.44 m) and infinity.

2.3.1.2 Mirrors. The mirrors shall be 0.125 inch (3.18 mm) thick and all edges shall be ground. Mirrors shall be provided with easily attachable, fungus resistant, soft covers. Mirrors shall be mounted in such a way as to permit easy removal or replacement. The right and left mirrors shall be interchangeable.

2.3.1.3 Prisms. The prisms shall be high-grade, right angle, 45-degree prisms, free from color and defects that would impair vision in normal use. The ends shall be fine ground

2.3.2 Binoculars. The binoculars shall be constructed such that, in combination with the basic stereoscope, a magnification of approximately four-by power and a resolution of at least 358.98 lines per inch (14 lines per mm) are obtained. Each field of view shall be not less than 1.625 inches (4.13 cm) in diameter at the plane of the photographs. All glass surfaces shall be coated with a reflection reducing film. The image presented shall be upright. The binoculars shall accommodate varying interpupilary distances and shall be adjustable to ± 5 diopters. The binoculars shall be mounted on the steroscope so that they can be easily swung into or out of position for viewing without being detached. The mounting shall permit photographs properly oriented for viewing with binoculars to be viewed without them. When in the viewing position, the axes of the binoculars shall be coincident with the lens axes of the basic stereoscope.

2.3.3 Case. The stereoscope and binoculars shall be furnished with a fitted, fungus resistant case that will protect the instrument from damage during transport.

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3. QUALITY ASSURANCE PROVISIONS. The supplier shall comply with the following for certification and acceptability.

3.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

3.2 Market acceptability. The stereoscope offered shall be of the latest model of the standard product of the supplier and shall have been used in the field for at least one year by commercial organizations.

3.3 Examination. Each stereoscope shall be examined for compliance with the requirements specified in 2.2 through 2.3.3. Any modification necessary following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects lessening maximum efficiency shall constitute cause for rejection.

4. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

5. NOTES.

5.1 The procuring agency should specify the level of preservation, packing, and marking required.

5.2 When this Commercial Item Description is used for procurement the contractor certification clause should appear in the solicitation.

5.3 Recovered materials. For the purpose of this requirement, recovered materials are those which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. Used, rebuilt, or remanufactured component, pieces and parts shall not be incorporated in the stereoscope.

5.4 Subject term (key word) listing.

photography
viewer

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Custodians:
Army - CE
Air Force - 99

Preparing Activity
Army - CE
(Project 6675-0029)

Review Activity:
DLA- GS

Civil Agency Coordinating Activity:
GSA - FSS

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